

Shortening of Cardiac Action Potential Duration Near an Insulating Boundary

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ABSTRACT

It is known, from both experiments and simulations, that cardiac action potentials are shortened near a nonconducting boundary. In the present paper, this effect is studied in a simple, two-current ionic model, with propagation restricted to a one-dimensional fiber. An asymptotic approximation for the dependence of action potential duration on distance to the boundary is derived. This estimate agrees well with simulations, and it predicts that the effect occurs in a layer much wider than the usual space constant, which is consistent with experimental observations.

Keywords: action potential duration; insulating boundary; one-dimensional cable.

1 Introduction

Cardiac cells are *excitable* in the sense that a brief electrical stimulus can cause the transmembrane voltage v to experience a prolonged elevation known as an *action potential* (see Figure 3a). Action potentials propagate through tissue because neighboring cells are connected electrically via intercellular channels (called gap junctions). These propagating electrical waves coordinate the contraction of cardiac muscle as needed for effective pumping of blood. Understanding the timing of these waves is a first step towards detecting and preventing potentially fatal arrhythmias.

As usual, let us define the *action potential duration (APD)* as the amount of time in which the voltage remains elevated above a specified threshold v_{thr} . In experiments, the threshold voltage is chosen anywhere from 70 to 90 percent¹ recovery from the peak voltage to the resting voltage.

¹In this study, our results are not especially sensitive to the choice of v_{thr} .

In a recent *in vitro* experiment [6], a small piece of frog heart tissue was stimulated using a unipolar pacing electrode and APD was measured on the surface of the tissue with voltage-sensitive fluorescent dyes. Figure 1 shows the APD's measured along a representative radial line emerging from the stimulus site. As the figure shows, APD values near the electrode were 10% larger than values at the edge of the tissue. To prove that this effect was not simply the result of tissue inhomogeneity, the stimulating electrode was placed at several different sites around the tissue; always APD exhibited the same shortening with increasing distance from the electrode.

Although it is not apparent from an experiment in such a small sample, this dependence of APD on distance from the stimulus site is a boundary effect. We support this claim with simulations using the two-current ionic model of [7, 9], which will be described in Section 2. In the simulations shown in Figure 2a, on a cable 1 cm long, APD decreases approximately linearly with distance from the stimulus site. By contrast, in those of Figure 2b, on a cable 4 cm long, APD is independent of position away from the ends of the cable, is elevated locally near the stimulus end, and is depressed locally near the distal end.

In this paper we study how the duration of a single action potential propagating along a long fiber is changed near the end of the fiber. In order to avoid modeling the details of stimulation, we restrict our attention to the distal end of the fiber. Specifically, for the two-current model, we derive an asymptotic approximation for $\delta APD(x)$, the change in APD expressed as a function of distance x along the fiber from a non-conducting boundary. This analytical result for a simplified model sheds light on phenomena in more realistic models or experiments; for example, that the APD is shortened in a surprisingly wide layer—several times larger than the space constant [8, 12] of the tissue.

Although this shortening is a relatively small fraction of the total APD, such boundary effects may play a decisive role in generating arrhythmias. For example, under rapid periodic pacing, tissue may bifurcate into a period-2 orbit called *alternans*: i.e., the response to identically spaced stimuli alternates between short and long APD's. In both simulations and experiments, as the pacing rate increases, alternans appear first and are strongest close to a boundary [5, 11].

The remainder of the paper is organized as follows. In Section 2, we recall the two-current ionic model [7, 9], including asymptotic estimates for APD in this model and the description of traveling waves in it. In Section 3, we derive our main result: an approximate formula for $\delta APD(x)$ near an insulating boundary. Finally, in Section 4 we show that our approximation agrees well with simulations, and we characterize how this phenomenon depends on the parameters in the model.

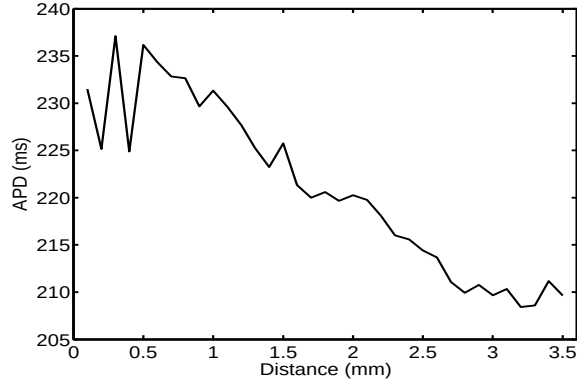


Figure 1: Experimentally measured spatial variation of APD. Stimuli were applied at $x = 0$, where x denotes distance from the pacing electrode.

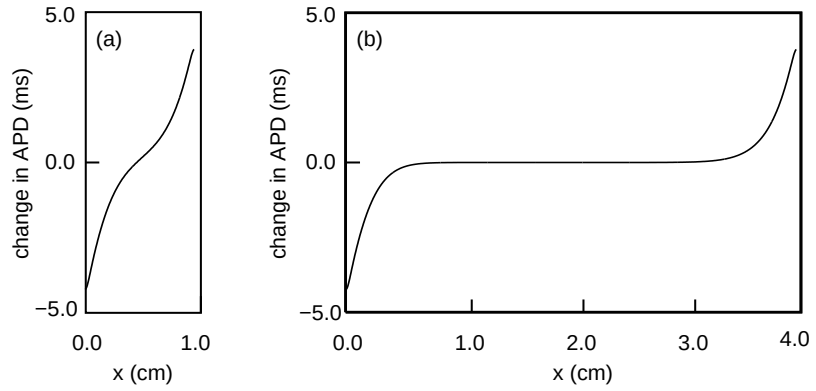


Figure 2: Spatial variation of APD in (a) a 1.0-cm cable and (b) a 4.0-cm cable as generated by numerical simulation of (4–5) with parameters from Table 1. Stimuli were applied at the *right* end of the fiber.

2 The two-current model

2.1 The equations

The two-current model [7, 9] for a patch² of tissue consists of two ordinary differential equations (ODEs) for two variables: the transmembrane voltage $v(t)$ and an inactivation gate variable $h(t)$, both of which are dimensionless and scaled to vary between 0 and 1. The voltage equation takes the form

$$\frac{dv}{dt} = F(v, h) = J_{\text{in}}(v, h) + J_{\text{out}}(v), \quad (1)$$

where $F(v, h)$ is a sum of inward and outward currents given by

$$J_{\text{in}}(v, h) = \frac{h}{\tau_{\text{in}}} v^2 (1 - v), \quad J_{\text{out}}(v) = -\frac{v}{\tau_{\text{out}}}. \quad (2)$$

The parameters τ_{in} and τ_{out} are time constants that specify the strength of the inward and outward currents, respectively. The other variable, the gate that regulates inward current flow, obeys the ODE

$$\frac{dh}{dt} = G(v, h) = \begin{cases} (1 - h)/\tau_{\text{open}}, & v < v_{\text{crit}} \\ -h/\tau_{\text{close}}, & v > v_{\text{crit}}. \end{cases} \quad (3)$$

Representative values for the time constants in (2,3) are given in Table 1.

Equations (1, 3) are easily modified to model the propagation of action potentials in a homogeneous one-dimensional cable—that is, an idealized fiber of cells electrically coupled via gap junctions. If x denotes distance along the fiber, the voltage $v = v(x, t)$ obeys a partial differential equation (PDE)

$$\frac{\partial v}{\partial t} = \kappa \frac{\partial^2 v}{\partial x^2} + F(v, h), \quad (4)$$

where κ is a diffusion constant; and the gate variable satisfies

$$\frac{\partial h}{\partial t} = G(v, h), \quad (5)$$

which by contrast contains no x -derivatives. In a finite-length or semi-infinite cable, Neumann boundary conditions on the voltage, corresponding to no axial current, are imposed at the ends of the cable.

When numerical simulations are required, we will use the parameter values listed in Table 1. Note that these time constants satisfy

$$\tau_{\text{in}} \ll \tau_{\text{out}} \ll \tau_{\text{open}}, \tau_{\text{close}}; \quad (6)$$

below we shall always assume the various time scales are well separated in this way.

²That is, a piece of tissue so small that spatial extent is negligible.

Parameter	Meaning	Units	Value
τ_{in}	time constant	ms	0.2
τ_{out}	time constant	ms	10
τ_{open}	time constant	ms	130
τ_{close}	time constant	ms	150
v_{crit}	critical voltage	dimensionless	0.13
κ	diffusion constant	cm^2/ms	0.001

Table 1: Sample parameter values for the two-current model.

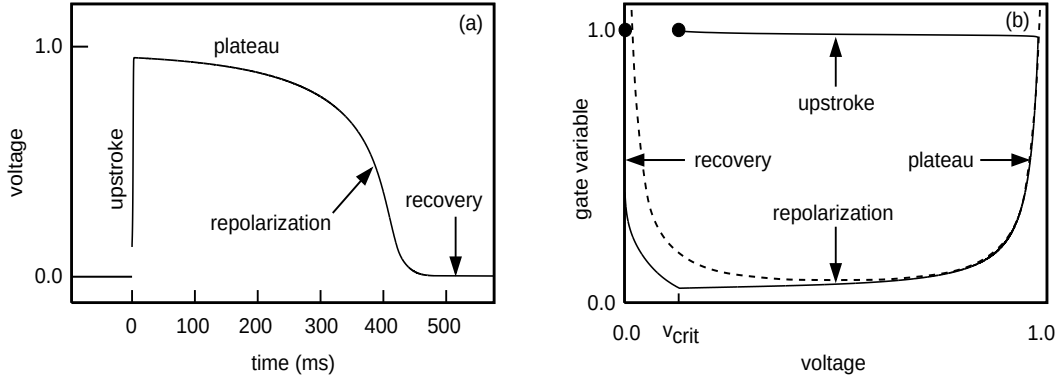


Figure 3: (a) Voltage trace illustrating the four phases of the action potential. (b) The corresponding phase plane trajectory. The dashed curve is the graph of the nullcline (7). Both figures were generated by numerical solution of (1–3) using the parameters in Table 1 and $v(0) = 0.13$, $h(0) = 1.0$.

2.2 Asymptotic estimates for APD in a patch

Observe that $(v, h) = (0, 1)$ is a stable equilibrium of the ODEs (1,3). These equations exhibit excitability in that, if a stimulus raises the voltage slightly above this equilibrium value, then the voltage increases further and remains elevated for a long period before returning to equilibrium, as sketched in Figure 3. The structure of such an action potential may be characterized with asymptotics using the fact that, because of (6), the voltage evolves much more rapidly than the gate unless (v, h) lies close to the nullcline $\{dv/dt = 0\}$. By (1,2), on this nullcline either $v = 0$ or

$$h = \frac{\tau_{\text{in}}}{\tau_{\text{out}}} \frac{1}{v(1-v)}. \quad (7)$$

Alternatively, solving (7) for v , we obtain

$$v = v_{\pm}(h) = \frac{1}{2} \left(1 \pm \sqrt{1 - \frac{h_{\text{min}}}{h}} \right), \quad (8)$$

where

$$h_{\text{min}} = \frac{4\tau_{\text{in}}}{\tau_{\text{out}}}. \quad (9)$$

Phase	Name	Duration (order of mag.)	Simplification	Dominant equation
1	Upstroke	τ_{in}	$h \approx \text{Const}$	(1)
2	Plateau	τ_{close}	$J_{\text{in}} + J_{\text{out}} \approx 0$	(3)
3	Repolarization	$\tau_{\text{close}}^{1/3} \tau_{\text{out}}^{2/3}$	$h \approx \text{Const}$	(1)
4	Recovery	τ_{open}	$v \approx 0$	(3)

Table 2: Summary of asymptotics during the four phases of an action potential

As illustrated in Figure 3, an action potential has four distinct phases. The dominant behavior in each phase may be described as follows and as summarized in Table 2. (See [9] for a more detailed discussion of the asymptotics.)

1. *Upstroke phase*: Following a successful stimulus, the strong inward current J_{in} raises the voltage until it reaches $V_+ = v_+(1)$ on the right branch of the nullcline (8). This occurs in a time on the order of $\mathcal{O}(\tau_{\text{in}})$.
2. *Plateau phase*: As the gate closes according to (3)—i.e., $h(t) = \exp(-t/\tau_{\text{close}})$ —the voltage follows the nullcline (8), keeping the inward and outward currents balanced.
3. *Repolarization phase*: When the gating variable reaches h_{min} given by (9), the solution “falls off the nullcline”. I.e., at this point the inward current is no longer able to balance

the outward current, and the voltage drops toward zero in a time that is short³ compared to τ_{close} .

4. *Recovery phase, or diastole:* The voltage stays small and the gate reopens with a time constant τ_{open} .

In Section 4.2 it will be useful to have estimates for the duration of such an action potential, which we denote by APD_{max} since this action potential is the largest possible APD that a cell can exhibit⁴. An APD consists of all of phase 2 plus parts of phases 1 and 3. According to (6), phases 1 and 3 are much shorter than phase 2. In fact, in leading-order asymptotics, APD_{max} equals the time in phase 2 during which, starting from 1, the gate decays exponentially to h_{min} . Thus,

$$\text{APD}_{\text{max}} \sim \tau_{\text{close}} \ln \left(\frac{1}{h_{\text{min}}} \right). \quad (10)$$

Strictly speaking, to compute APD we should introduce a threshold voltage v_{thr} and define APD as the length of time that $v(t) > v_{\text{thr}}$. However, because the voltage evolves rapidly in phases 1 and 3, the low-order estimates for APD_{max} do not depend on v_{thr} . Nevertheless, the choice of v_{thr} does affect (slightly) the boundary corrections to APD computed below.

At the next order of asymptotics, phase 3 *does* contribute to APD_{max} . At first glance, one might guess that the duration of phase 3 is $\mathcal{O}(\tau_{\text{out}})$ since this parameter characterizes the strength of J_{out} , the dominant current in this phase. In terms of the small parameter

$$\varepsilon = \frac{\tau_{\text{out}}}{\tau_{\text{close}}}, \quad (11)$$

this would represent an $\mathcal{O}(\varepsilon)$ correction to (10). However, at the start of phase 3, pending further decay of h and hence of J_{in} , the two currents nearly cancel one another. As shown in [2], this effect slows the decrease of the voltage and thus lengthens the APD. (Cf. The analysis of Cartwright [3] and Dorodnitsyn [4] concerning relaxation oscillations in the van der Pol equation.) Specifically, we find that (10) is augmented to

$$\text{APD}_{\text{max}} \sim \tau_{\text{close}} \left\{ \ln \left(\frac{1}{h_{\text{min}}} \right) + z_1 \varepsilon^{2/3} \right\}, \quad (12)$$

where $z_1 = 2.33811 \dots$ is the first zero of the Airy function, $\text{Ai}(-z)$. The next-order contribution beyond (12) of phase 3 to APD is $\mathcal{O}(\varepsilon \ln \varepsilon)$, and the entire contribution of phase 1 is of even higher order. Although for typical parameter values, this next order is comparable to $\mathcal{O}(\varepsilon^{2/3})$, formula (12) predicts APD_{max} with surprisingly high accuracy. For example,

³As asserted in Equation (12) below, the duration of this phase is $\mathcal{O}(\tau_{\text{close}}^{1/3} \tau_{\text{out}}^{2/3})$.

⁴ APD_{max} refers to an action potential produced by a stimulus following a long diastolic interval. Under periodic pacing, shorter action potentials are produced, especially under rapid pacing (which is not considered here).

using the parameters in Table 1, we estimate from formula (12) that $\text{APD}_{\max} = 436.5$ ms, less than 1% below the true value (obtained numerically) of 439.3 ms. By comparison, the leading-order estimate (10) predicts that $\text{APD}_{\max} = 378.7$ ms, a relative error of 14%.

2.3 Traveling-wave solutions in an infinite fiber

We seek a traveling-wave solution of Equations (4, 5),

$$v(x, t) = V(t + x/c), \quad h(x, t) = H(t + x/c), \quad (13)$$

where the speed c is to be determined. This yields a system of two ODEs for V and H with independent variable $\xi = t + x/c$. We shall solve this system approximately in terms of matched asymptotics. Specifically, for $\xi < 0$ the solution we find is quiescent ($V \equiv 0, H \equiv 1$); for $\xi > 0$ it is well-approximated by phases 2,3, and 4 of an action potential in a patch model; and these two outer regions are separated by an inner or wave-front layer in which voltage diffusion is crucial (see Figure 4).

In the inner layer, in a time that is $\mathcal{O}(\tau_{\text{in}})$, the voltage climbs from 0 to V_+ , where (cf. (8))

$$V_{\pm} = v_{\pm}(1) = \frac{1}{2} \left(1 \pm \sqrt{1 - h_{\min}} \right). \quad (14)$$

We may approximate $H \equiv 1$ in the wave-front layer because, by Equation (6), the time constants regulating H are much larger than τ_{in} . In terms of a scaled inner coordinate $\tilde{\xi} = \tau_{\text{in}}^{-1}(t + x/c)$, to leading order the inner solution solves

$$V' = \frac{1}{c^2} \frac{\kappa}{\tau_{\text{in}}} V'' + \tau_{\text{in}} F(V, 1), \quad (15)$$

where prime indicates differentiation with respect to $\tilde{\xi}$. Equation (15) is subject to boundary conditions

$$\lim_{\tilde{\xi} \rightarrow -\infty} V(\tilde{\xi}) = 0, \quad \lim_{\tilde{\xi} \rightarrow \infty} V(\tilde{\xi}) = V_+. \quad (16)$$

Proposition 2.1. *Problem (15,16) has a solution if*

$$c = \left(\frac{1}{2} V_+ - V_- \right) \sqrt{\frac{2\kappa}{\tau_{\text{in}}}}. \quad (17)$$

Proof. We seek a solution of (15) using a common trick (see, for example, [10]): Solutions of the simpler ODE

$$V' = -aV(V - V_+), \quad (a > 0) \quad (18)$$

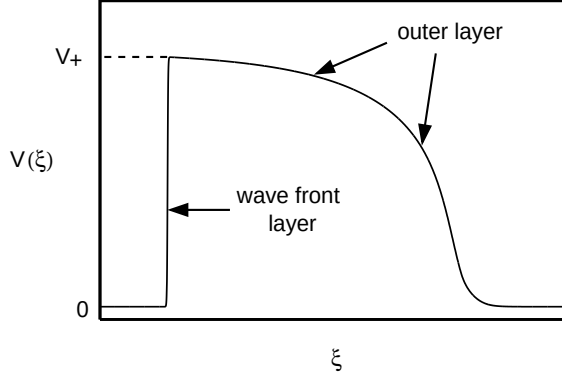


Figure 4: Voltage profile of a traveling pulse moving from right to left on an infinite cable.

are also solutions of (15) provided a and c are chosen appropriately. Indeed, substituting (18) into (15) leads to a system of two equations with two unknowns a and c , the solution of which yields

$$a = \frac{\frac{1}{2}V_+ - V_-}{\tau_{\text{in}}} \quad (19)$$

and c given by (17). (See [1] for more detail.) $\square$

Incidentally, (17) predicts $c = 46.9$ cm/sec, while we find $c = 46.7$ cm/sec from numerical solution of (4,5) using the parameter values in Table 1.

In the outer region $\xi > 0$, let $V(\xi), H(\xi)$ solve the ODE

$$\frac{dV}{d\xi} = F(V, H) \quad \frac{dH}{d\xi} = G(V, H),$$

subject to initial conditions

$$V(0) = V_+ \quad H(0) = 1. \quad (20)$$

This corresponds to phases 2, 3, and 4 of an action potential in a patch. We claim that substitution of $V(\xi), H(\xi)$ into (13) gives a leading-order solution $v(x, t), h(x, t)$ of the PDE (4,5). Consider for example values of ξ corresponding to phase 2 of an action potential. In this range of ξ

$$\frac{dV}{d\xi} \sim \frac{1}{\tau_{\text{close}}}, \quad \frac{d^2V}{d\xi^2} \sim \frac{1}{\tau_{\text{close}}^2}.$$

Thus, substituting into (13) and recalling (17), we see that

$$\kappa \frac{\partial^2 v}{\partial x^2} \sim \frac{\tau_{\text{in}}}{\tau_{\text{close}}^2} \sim \frac{\tau_{\text{in}}}{\tau_{\text{close}}} \frac{\partial v}{\partial t}.$$

Hence, since $\tau_{\text{in}}/\tau_{\text{close}} \ll 1$, the diffusion term, which distinguishes the PDE (4) from the ODE (1), is of higher order. Similarly for phases 3 and 4, which proves the claim. Comparing (16) and (20), we see that the outer solution matches continuously onto the inner solution. This completes the asymptotic construction of the traveling-wave solution of (4,5).

We summarize: To leading order, the traveling wave differs from an action potential on a patch only in the wave-front layer. In this layer, the speed of the traveling wave is determined by the balance between diffusion and the strength of the inward current.

3 A formula for the boundary correction to APD

3.1 The governing equations on a half-line

Consider a solitary action potential propagating along a long, homogeneous fiber toward an insulating boundary at $x = 0$. The dynamics can be modeled using (4, 5) on a half-space $\{0 < x < \infty, -\infty < t < \infty\}$ subject to the boundary condition

$$\frac{\partial v}{\partial x}(0, t) = 0 \quad (-\infty < t < \infty) \quad (21)$$

and the initial conditions

$$\lim_{t \rightarrow -\infty} \max_{0 \leq x < \infty} |v(x, t) - V(t + x/c)| = 0 \quad (22)$$

$$\lim_{t \rightarrow -\infty} \max_{0 \leq x < \infty} |h(x, t) - H(t + x/c)| = 0. \quad (23)$$

In words, the initial conditions require that the difference between the exact solution and a pure traveling-wave solution is negligible when the pulse is far from the boundary.

We seek a solution of this problem of the form

$$v(x, t) = V(t + x/c) + w(x, t), \quad h(x, t) = H(t + x/c), \quad (24)$$

where V, H is the traveling wave solution constructed in Section 2.3 and w is a small correction. (We neglect the correction to the gate variable, which is of higher order since the gate evolves more slowly than the voltage.) Substituting (24) into (4,21,22) and gathering the leading-order terms in w , we obtain the linearized problem

$$\frac{\partial w}{\partial t} = \kappa \frac{\partial^2 w}{\partial x^2} + \gamma(x, t)w \quad (x > 0, -\infty < t < \infty) \quad (25)$$

$$\frac{\partial w}{\partial x}(0, t) = -\frac{1}{c}V'(t) \quad (-\infty < t < \infty) \quad (26)$$

$$\lim_{t \rightarrow -\infty} w(\cdot, t) = 0, \quad (27)$$

where

$$\gamma(x, t) = \frac{H(t + x/c)}{\tau_{\text{in}}} [2V(t + x/c) - 3V(t + x/c)^2] - \frac{1}{\tau_{\text{out}}}. \quad (28)$$

Below we shall solve (25–27) making the approximation $\gamma(x, t) \approx \gamma(0, t)$: i.e.,

$$\gamma(x, t) \approx \gamma(t) = \frac{H(t)}{\tau_{\text{in}}} [2V(t) - 3V(t)^2] - \frac{1}{\tau_{\text{out}}}. \quad (29)$$

This approximation will be justified in Section 4.2.

3.2 Solution of the linearized problem

As a first step in attacking (25–27), we solve an analogous problem for the heat equation. Specifically, consider

$$\frac{\partial w}{\partial t} = \kappa \frac{\partial^2 w}{\partial x^2} \quad (x > 0, \quad -\infty < t < \infty) \quad (30)$$

$$\kappa \frac{\partial w}{\partial x}(0, t) = -j(t) \quad (-\infty < t < \infty) \quad (31)$$

$$\lim_{t \rightarrow -\infty} w(x, t) = 0 \quad (x > 0) \quad (32)$$

where in (31) we assume that j is bounded and integrable. The minus sign in (31) reflects the fact that a net influx of heat corresponds to a negative temperature gradient. We will show that the solution of (30–32) is given by

$$w(x, t) = \int_0^\infty K(x, s) j(t - s) ds, \quad (33)$$

where

$$K(x, t) = \sqrt{\frac{1}{\pi \kappa t}} \exp\left(-\frac{x^2}{4\kappa t}\right). \quad (34)$$

Note that (34) differs from the usual heat kernel by a factor of 2, a consequence of our working on a *half*-line.

One may derive (33) by (i) defining $\tilde{w}(x, t) = w(x, t) - xj(t)$, which satisfies a homogeneous Neumann boundary condition but an inhomogeneous equation, (ii) solving for $\tilde{w}$ by making an even extension to the whole line, and (iii) manipulating the resulting solution, including an integration by parts. However, we prefer the following more intuitive verification of (33).

We begin with a weak reformulation of the boundary condition (31). Suppose that w is a smooth solution of (30,31) that decays as $x \rightarrow \infty$. We claim that for all $t_1, t_2 \in \mathbf{R}$

$$\int_0^\infty [w(x, t_2) - w(x, t_1)] dx = \int_{t_1}^{t_2} j(t) dt. \quad (35)$$

The claim may be proved by rewriting the LHS of (35) as

$$\int_0^\infty dx \int_{t_1}^{t_2} dt \frac{\partial w}{\partial t}(x, t),$$

interchanging the order of integration, invoking the PDE (30), applying the fundamental theorem of calculus to $\int_0^\infty \partial^2 w / \partial x^2 dx$, and making use of (31).

Let us extend K to negative times (still for $x \geq 0$) by

$$\tilde{K}(x, t) = \begin{cases} K(x, t) & \text{if } t > 0 \\ 0 & \text{if } t \leq 0, \text{ provided } (x, t) \neq (0, 0). \end{cases} \quad (36)$$

Although $\tilde{K}$ is undefined at $(0, 0)$, it is locally integrable on $[0, \infty) \times \mathbf{R}$, so it defines a distribution. Of course $\tilde{K}$ satisfies the heat equation and the initial condition (32). It is easily checked that

$$\int_0^\infty \tilde{K}(x, t) dx = \begin{cases} 1 & \text{if } t > 0 \\ 0 & \text{if } t < 0. \end{cases}$$

Thus, in the weak sense of (35), $\tilde{K}$ satisfies the boundary condition

$$\kappa \frac{\partial \tilde{K}}{\partial x}(0, t) = -\delta(t),$$

where $\delta(t)$ is the Dirac delta function. Equation (33) therefore follows by convolution.

If the coefficient γ in (25) depends only on t , then we may reduce this equation to the heat equation by introducing an integrating factor. Specifically, let

$$\Gamma(t) = \int_t^{\text{APD}_{\max}} \gamma(s) ds,$$

the upper limit being chosen for later convenience. Then (25) may be rewritten as

$$\frac{\partial}{\partial t}[e^{\Gamma(t)} w] = \kappa \frac{\partial^2}{\partial x^2}[e^{\Gamma(t)} w].$$

Applying (33), we find that the solution of (25–27) is given by

$$w(x, t) = \frac{\kappa}{c} \int_0^\infty K(x, s) e^{[\Gamma(t-s) - \Gamma(t)]} V'(t-s) ds. \quad (37)$$

3.3 The boundary correction to APD

We apply formula (37) to extract the spatial variation in APD in the vicinity of the insulating boundary. Given $x \geq 0$, we wish to estimate the amount of time in which the voltage $v(x, t)$ remains elevated above the threshold voltage v_{thr} . To do so, we must determine the two times t for which

$$v(x, t) = V\left(t + \frac{x}{c}\right) + w(x, t) = v_{\text{thr}} \quad (x \geq 0 \text{ fixed}); \quad (38)$$

i.e., the arrival times at x of the wave front and of the wave back.

In the asymptotic construction of the traveling wave in Section 2.3, the wave front of the unperturbed solitary pulse reaches the boundary $\{x = 0\}$ at time $t = 0$; and the wave back at $t = \text{APD}_{\text{max}}$. In other words,

$$V(0) = V(\text{APD}_{\text{max}}) = v_{\text{thr}}. \quad (39)$$

Using (39) to generate a first estimate, we seek approximate times when (38) is satisfied by assuming

$$t + x/c = \delta_1(x), \quad \text{or} \quad t + x/c = \text{APD}_{\text{max}} + \delta_2(x), \quad (40)$$

where $\delta_i(x)$, representing the effect of the perturbing term w in (38), is small. Expanding V about 0 or APD_{max} , invoking (39), and neglecting $\delta_i(x)$ in the argument of w , we simplify (38) to

$$V'(0)\delta_1(x) + w(x, -x/c) = 0, \quad \text{or} \quad V'(\text{APD}_{\text{max}})\delta_2(x) + w(x, \text{APD}_{\text{max}} - x/c) = 0.$$

Solving these linear equations for $\delta_i(x)$, we compute that the *change* in APD due to pulse interaction with the boundary is given by

$$\delta \text{APD}(x) \equiv \delta_2(x) - \delta_1(x) \approx -\frac{w(x, \text{APD}_{\text{max}} - x/c)}{V'(\text{APD}_{\text{max}})} + \frac{w(x, -x/c)}{V'(0)}. \quad (41)$$

Since $V'(0) = \mathcal{O}(\tau_{\text{in}}^{-1})$ represents the steep slope (see Figure 4) of the wavefront of the undisturbed traveling pulse, the second term in (41) makes only a higher-order contribution to δAPD . Retaining only the first term in (41) and recalling (37), we find our main estimate

$$\delta \text{APD}(x) \approx -\frac{\kappa}{c} \int_0^\infty K(x, s) e^{\Gamma(\text{APD}_{\text{max}} - x/c - s) - \Gamma(\text{APD}_{\text{max}} - x/c)} \frac{V'(\text{APD}_{\text{max}} - x/c - s)}{V'(\text{APD}_{\text{max}})} ds. \quad (42)$$

4 Discussion and conclusions

4.1 Numerical simulations

Computer simulations show that the shortening of APD predicted by (42) is in good quantitative agreement with results from numerical solutions of (4–5). Specifically, Figure 5 shows a comparison of the actual change in APD (solid curve) with the change predicted by (42) (dashed curve). The solid curve was generated by numerical simulation of (4–5) using the parameter values in Table 1, a threshold voltage $v_{\text{thr}} = v_{\text{crit}} = 0.13$, and a cable length of 4 cm. Initially, the cable was at rest ($v = 0$ and $h = 1$ for all cells). Then, voltage was suddenly raised to $v = 0.8$ for all cells with $3.95 \leq x \leq 4.0$, eliciting an action potential propagating from right to left. APD measurements were taken at all cells outside the stimulus region, and the plot shows $\delta\text{APD}(x)$ in the region within 1 cm of the distal end of the cable. In generating the dashed curve using formula (42), Γ and V' were computed by numerical simulation of the traveling wave.

4.2 Order of magnitude estimates

(a) *Intuitive discussion*

In this section we estimate the order of magnitude of $\delta\text{APD}(x)$ in (42) in order to extract the dependence of the correction on the parameters in the model: i.e., on the time constants $\tau_{\text{in}}, \tau_{\text{out}}, \tau_{\text{close}}, \tau_{\text{open}}$. Some intuition about how diffusive currents shorten the APD near a boundary facilitates the discussion. Since the main effect occurs during repolarization, we consider a representative cell as this phase of the traveling wave passes through it, as indicated in Figure 6. For both cases shown in the figure, the diffusive current $-\kappa\partial v/\partial x$ moves charge from this cell to its neighbors on the right, where the action potential has progressed further. Far from the boundary, as in Figure 6a, the loss of charge to neighbors on the right is approximately canceled by the gain of charge from neighbors on the left. By contrast, near the boundary, as in Figure 6b, neighbors on the left are missing. Thus, in the latter case the diffusive current lowers the voltage in the wave-back region and hence shortens the APD.

Incidentally, the same effect near the wave front explains the discrepancy between the two graphs in Figure 5. The analytical result (42) neglects any effects near the wave front. However, near a boundary, the speed of an approaching pulse is accelerated because no diffusive current need be supplied to neighboring cells on the left, and this slightly lengthens APD. As may be seen in Figure 5, the effect near the wave front is smaller and more localized.

(b) *Preliminary approximations*

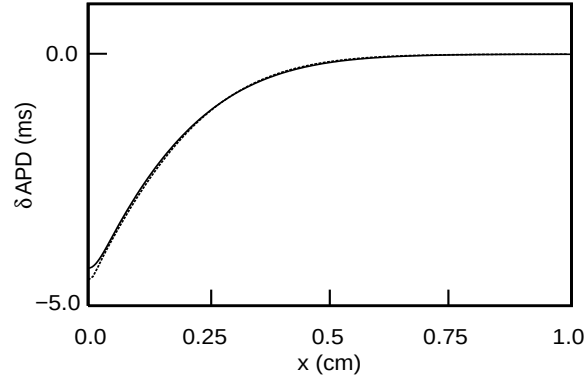


Figure 5: Comparison of $\delta\text{APD}(x)$ as predicted by (42) (dashed curve) and by numerical solution of the original PDEs (solid curve).

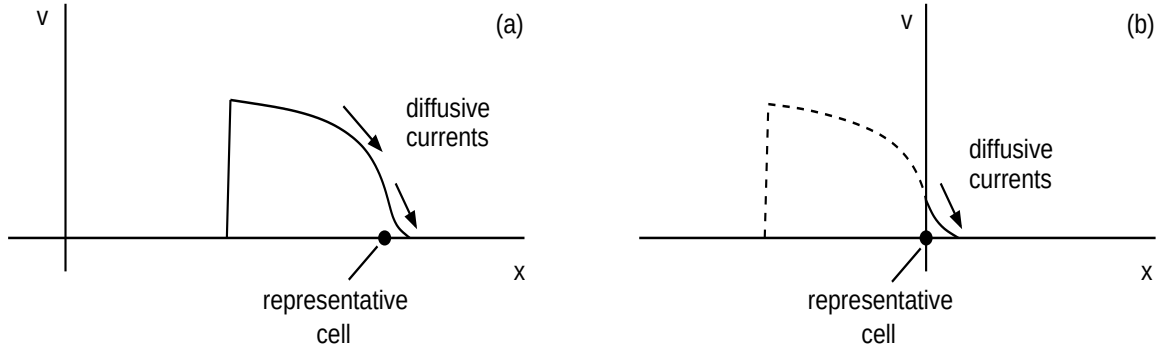


Figure 6: Diffusive currents for a pulse (a) far from an insulating boundary and (b) after reaching the insulating boundary.

The integrand in (42) may be decomposed into three factors,

$$K(x, s), \quad e^{\Gamma(\text{APD}_{\max} - x/c - s) - \Gamma(\text{APD}_{\max} - x/c)}, \quad \text{and} \quad \frac{V'(\text{APD}_{\max} - x/c - s)}{V'(\text{APD}_{\max})}.$$

Regarding the third factor, the numerator $V'(t)$ is largest for times in the repolarization phase (see Figure 7). According to (12), the correction to $\text{APD}_{\max}$ in leading-order asymptotics, this phase has width approximately

$$W = \tau_{\text{close}} z_1 \varepsilon^{2/3}. \quad (43)$$

Therefore, observing that the third factor above equals 1 when both $x = 0$ and $s = 0$, we propose the crude approximation,

$$\frac{V'(\text{APD}_{\max} - x/c - s)}{V'(\text{APD}_{\max})} \approx \begin{cases} 1, & x/c + s \leq W \\ 0, & x/c + s > W. \end{cases} \quad (44)$$

However, as Figure 7 suggests, $V'(t)$ has a long tail extending to times well before the wave back region. Thus, the approximation by zero in (44) for this range may seem too severe. Nevertheless, as shown in Figure 8, the support of the second factor in (42), the exponential, is effectively confined to the wave back region. Moreover, in the wave back region this exponential is on the order of unity. Thus, (44) is an acceptable approximation for the *product* of the second and third factors in (42). Note that this approximation effectively limits the range of integration in (42) to the finite interval $0 \leq s \leq W - x/c$.

Regarding the first factor in (42), observe from (34) that if $x > 0$ is fixed and $s \rightarrow 0$, then $K(x, s)$ is exponentially small. On the other hand, the exponential factor in (34) tends to unity as $s \rightarrow \infty$. Based on this, we make the crude approximation

$$K(x, s) \approx \begin{cases} 0, & s \leq x^2/\kappa \\ 1/\sqrt{\pi\kappa s}, & s > x^2/\kappa. \end{cases} \quad (45)$$

This approximation also reduces the range of integration in (42) by imposing a nonzero lower limit.

(c) *The width of the boundary-correction layer*

Combining the approximations (44, 45), we see that the effective range of integration in (42) is the interval $(x^2/\kappa, W - x/c)$. This interval shrinks to naught if

$$\frac{x^2}{\kappa} = W - \frac{x}{c}. \quad (46)$$

In other words, (46) defines the width of the zone where $\delta \text{APD}(x)$ differs significantly from zero.

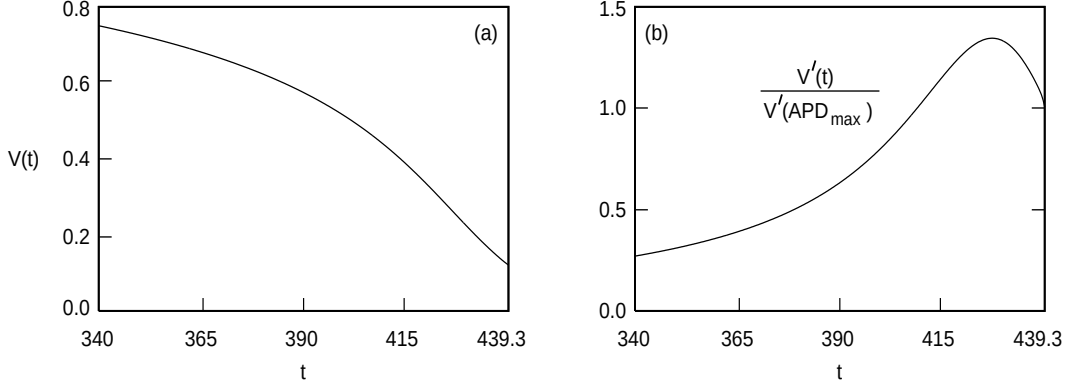


Figure 7: (a) $V(t)$ versus t . (b) $V'(t)/V'(APD_{\max})$ versus t . Both graphs are terminated at $t = APD_{\max}$

Let us show that, to lowest order, (46) is satisfied when

$$x \sim \varepsilon^{-1/6} \sqrt{\kappa \tau_{\text{out}}}. \quad (47)$$

To prove this, first observe that (47) comes from solving $x^2/\kappa = W$: i.e., neglecting the term x/c on the RHS of (46). For the value of x given by (47), after some rearranging of powers of the time constants, we see that this neglected term satisfies

$$\frac{x}{c} \sim \sqrt{\frac{\tau_{\text{in}}}{\tau_{\text{out}}}} \varepsilon^{5/6} \tau_{\text{close}},$$

which is much smaller than W and so can be neglected in solving (46).

Thus, (47) estimates the width of the boundary-correction layer. Figure 9 shows that the $\varepsilon^{-1/6}$ scaling for the width of the boundary layer is quite accurate, provided ε is not too large. Note that $\sqrt{\kappa \tau_{\text{out}}}$ in (47) is the space constant of the passive fiber [8, 12]. Thus, the boundary correction is significant in a layer much wider than the space constant.

We also give a less computationally intensive, heuristic derivation of (47). According to the discussion below (41), $\delta APD(x)$ is approximately proportional to $w(x, APD_{\max} - x/c)$, where w is the solution of (25–27). The effects of the inhomogeneous boundary condition (26) propagate into the interior of the domain according to the PDE (25). Now $V'(t)$, the inhomogeneity, differs significantly from zero only during the upstroke and repolarization phases. Regarding the former, we claim that its contribution to w at the end of the action potential is negligible. To see this, observe from (29) that, during the entire plateau phase, the coefficient $\gamma(t)$ of the zeroth-order term in (25) is negative and on the order of $\mathcal{O}(1/\tau_{\text{in}})$. Thus, because of this term, during the course of an action potential, the contribution of the upstroke phase will be reduced by a factor on the order of $\mathcal{O}(\exp(-\tau_{\text{close}}/\tau_{\text{in}}))$, which verifies the claim. By contrast, let us show that during repolarization the effect of γ is small. At the

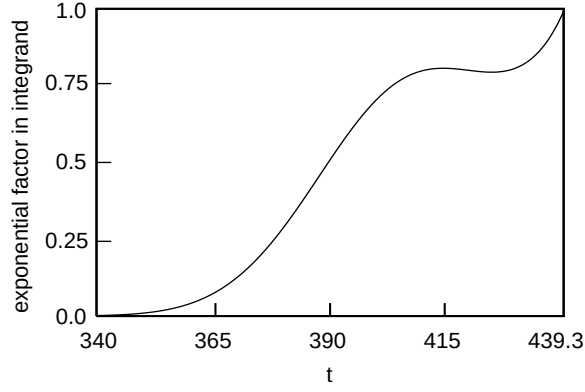


Figure 8: The exponential factor in the integrand of (42), with $x = 0$.

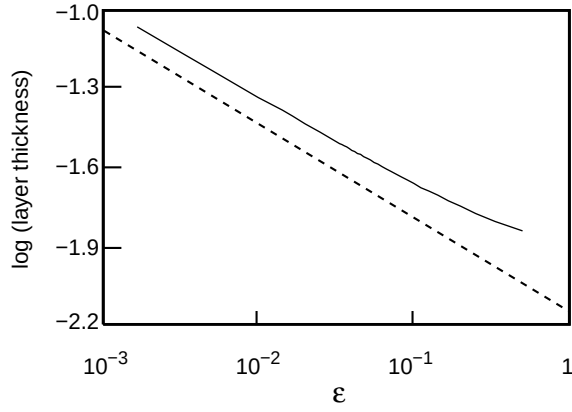


Figure 9: Log-log plot of layer thickness versus ϵ . Layer thickness was defined as the x value for which $(\text{APD}_{\text{mid}} - \text{APD}(x))/(\text{APD}_{\text{mid}} - \text{APD}(0)) = 1/e$, where APD_{mid} denotes the APD in the middle of the fiber. The dashed line has slope $-1/6$ and is included for reference.

start of repolarization $h = h_{\min} = 4\tau_{\text{in}}/\tau_{\text{out}}$. Substitution of this value of h into (29) would give

$$\gamma = \tau_{\text{out}}^{-1} [8V - 12V^2 - 1]. \quad (48)$$

This expression, which is much smaller than γ during the plateau phase, is even positive for $1/6 < V < 1/2$, typical values for the voltage during repolarization; and a positive γ would cause *growth* in w . However, (48) is an overestimate for γ since $h(t)$ continues to decay slowly during repolarization. In point of fact, during repolarization, $\gamma(t)$ is generally small and does become positive for a brief time. To a first approximation we neglect γ and regard w during this phase as satisfying the heat equation with *no* zeroth-order term. For the heat equation, by time $t = \text{APD}_{\max}$, the effects of a nonzero boundary condition for the period

$$\text{APD}_{\max} - W < t < \text{APD}_{\max}$$

will propagate a distance on the order of $\sqrt{\kappa W}$ into the interior. In other words, $w(x, \text{APD}_{\max})$ will become negligible when

$$x \sim \sqrt{\kappa W}; \quad (49)$$

moreover, since $\sqrt{\kappa W}/c \ll W$, the same conclusion holds for $w(x, \text{APD}_{\max} - x/c)$, the quantity needed in (41). On substitution of (43) into (49), this estimate for the width of the boundary layer reduces to (47).

Incidentally, the estimate (49), which is based on the duration of the repolarization phase, is perhaps more appropriate for generalization to sophisticated ionic models. In the present model, the duration of the repolarization phase and the time constant τ_{out} of the passive fiber are linked to one another, but in a more complex model they could well be independent of one another.

(d) *The maximum size of δAPD*

The correction $\delta \text{APD}(x)$ assumes its maximum at $x = 0$. Recalling (34) and substituting (44) for the product of the second and third factors in (42), we obtain

$$\delta \text{APD}(0) \approx -\frac{\kappa}{c} \int_0^W \frac{1}{\sqrt{\pi \kappa s}} ds. \quad (50)$$

Performing the integration in (50) and ignoring various dimensionless numerical factors like $\sqrt{\pi}$, we estimate

$$\delta \text{APD}(0) \sim -\frac{\kappa}{c} \sqrt{\frac{W}{\kappa}}. \quad (51)$$

Collecting powers of the parameters, we find

$$\delta \text{APD}(0) \sim -\sqrt{\frac{\tau_{\text{in}}}{\tau_{\text{out}}}} \varepsilon^{-1/3} \tau_{\text{out}}. \quad (52)$$

(e) *Justification of the approximation (29)*

The relation $\gamma(x, t) \approx \gamma(0, t)$ is based on the fact that $V(t+x/c), H(t+x/c)$, the traveling-wave solution, does not vary greatly over the range of x where w differs significantly from zero. This range of x was estimated in (47). During repolarization, the derivatives $dV/d\xi, dH/d\xi$ are of the order at most $\mathcal{O}(1/\tau_{\text{out}})$. Thus, in the worst case

$$|V(t+x/c) - V(t)| \sim \frac{1}{\tau_{\text{out}}} \frac{\varepsilon^{-1/6} \sqrt{\kappa \tau_{\text{out}}}}{c} \sim \varepsilon^{-1/6} \sqrt{\frac{\tau_{\text{in}}}{\tau_{\text{out}}}}. \quad (53)$$

and similarly for H . For the parameters in Table 1, the RHS of (53) is approximately 0.22, which is small but not extremely small. Thus, it is reassuring that, as illustrated in Figure 5, estimates based on the approximation (29) give highly accurate results.

4.3 Conclusions

In this paper we have studied how the duration of a propagating cardiac action potential is modified as it nears an insulating boundary. In order to obtain explicit analytical results, we described action potentials in a one-dimensional fiber using a simplified two-current model [7, 9]. Equation (42) estimates the shortening of APD at a site as a function of its distance from the boundary. As shown in Figure 5, this estimate agrees well with simulations. In (47) we determine how the width of the layer in which noticeable shortening occurs depends on the parameters in the model; and in (52), the maximum shortening. As is observed in experiments, the width of the layer is a large multiple of the space constant.

Acknowledgments

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